

Hanlin (Harry) Shi

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Biorobotics Lab, Robotics Institute, Carnegie Mellon University, Pittsburgh, PA 15213

EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Expected, May 2027

- Master of Science in Mechanical Engineering
- Cumulative GPA: 4.0/4.0

Bucknell University, Lewisburg, PA

May 2025

- Bachelor of Science in Mechanical Engineering, *summa cum laude*
- Minor in Mathematics
- Cumulative GPA: 3.94/4.0

HONORS AND AWARDS

Summa Cum Laude, Bucknell University

May 2025

- Awarded to graduating students in the highest band of cumulative grade point average

Member, Pennsylvania Eta Chapter of Tau Beta Pi Honor Society

November 2023

- A national engineering honor society founded in 1885 to recognize outstanding students who display both distinguished scholarship in technical fields and exemplary character

Dean's List, Bucknell University

Each completed semester at Bucknell

- Awarded to each student who achieves a 3.50 or higher GPA for the semester

PUBLICATIONS AND PRESENTATIONS

Journal Papers

1. **H. Shi**, J. Kallarackal, K. Karumanchi, S. Pellegrini, A. Orekhov, A. Vu, B. Vundurthy and H. Choset, "A unified compliant locomotion framework for snake robot navigation in complex external pipe networks," *IEEE Transactions on Automation Science and Engineering*, under review, 2026.

Conference Papers

1. **H. Shi** and W. L. Scott, "Crawling gait design for a caterpillar-inspired robot with tendon-driven flexible bending segments," in *7th IEEE-RAS International Conference on Soft Robotics*, 2024.

Presentations

- Lightning talk and poster, *7th IEEE-RAS International Conference on Soft Robotics*, 2024
- Poster, *13th Naval Academy Science and Engineering Conference*, 2024
- Posters, *22nd and 23rd Annual Kalman Research Symposium*, 2022 and 2023
- Posters, *Bucknell Engineering Student Research Symposium*, 2022 and 2023

RESEARCH EXPERIENCE

Graduate Researcher — Autonomous External Pipe Navigation for Snake Robots

August 2025 – Present

- *Biorobotics Lab, Robotics Institute, Carnegie Mellon University, Pittsburgh, PA*
- Research Advisor: Howie Choset, Ph.D.

The research focuses on closing the control and perception loops onboard a snake robot that climbs the exterior of pipe networks, regulating contact from the robot's own torque sensing and inferring the surrounding geometry from its own proprioception, for

applications in inspecting chemical processing plants, power generation facilities, and municipal water distribution pipelines whose complex geometries make inspection difficult and unsafe for humans.

- Developed the deviation-driven shape-based compliance (DSBC) controller, which treats only the deviation of measured joint torque from a reference as contact, halving traversal times across straight pipes, obstacles and junctions while lowering joint actuation effort
- Generalized the windowed rolling helix gait by expressing the head–tail phase offset as a function of the branch angle, so that a single formulation covers arbitrary junction geometries; validated on hardware at a 135° Y-junction
- Developed a real-time estimation pipeline built on the Virtual Chassis formalism that recovers the pipe axis, the robot's orientation and the pipe radius from joint encoders and IMUs alone, using a Levenberg–Marquardt centerline fit where the standard SVD estimator degenerates
- Corrected the radius model to account for the robot wrapping the pipe as a polygon rather than a circle, reducing diameter error from +21% to under 4%
- Estimated pipe inclination to approximately 2.5° mean error through a complementary filter fusing gyroscope and accelerometer measurements
- Reconstructed the pipe profile from gait-based odometry, integrating the known advance per rolling-helix cycle with the estimated axis direction and radius to recover changes in inclination and diameter without any external reference
- Currently developing a junction detection model that estimates contact force from joint torque, decomposing contact into a radial gripping component and an axial component that indicates a junction, and localizing where along the body that force acts

Undergraduate Researcher — Centipede-Inspired Soft Robot

August 2023 – May 2025

- *Department of Mechanical Engineering, Bucknell University, Lewisburg, PA*
- Research Advisor: William Scott, Ph.D.

The research focuses on developing a new design for a soft robot that mimics the crawling mechanism of a centipede to travel on uneven terrain for applications in interacting safely with humans and navigating complex environments that are difficult for humans to reach.

- Designed a modular segment that reproduces the centipede's coupling mechanically, so that one servo per segment drives a pair of flexible links that bend the body and raise the legs together, rather than coordinating separate actuators for body bending and leg lift
- Programmed an offboard controller that drives the segments through a traveling wave to produce the crawling gait
- Built the robot with modularity, so that gaits can be tested across different body lengths

Undergraduate Researcher — Sub-gram Aerial Robot Simulation

May 2024 – August 2024

- *Soft and Micro Robotics Lab, Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology, Cambridge, MA*
- Research Advisor: Yufeng (Kevin) Chen, Ph.D.

The research focuses on using numerical methods to develop a simulation for the kinematics of a sub-gram aerial robot flapping-wing motion to predict the lift force required to be generated by the Dielectric Elastomer Actuator of the robot, for applications in demonstrating how soft actuators enable controlled flight and illustrating the potential of developing next-generation agile soft robots.

- Developed a MATLAB motion simulation model predicting the position of the flapping wing at each time step due to active stroke motion and passive aerodynamic pitching about a compliant hinge
- Predicted the state variables of any arbitrary position on the wing, and the aerodynamic forces generated by the wing over a stroke
- Generated an animation to illustrate the motion of the flapping wing

Undergraduate Researcher — Caterpillar-Inspired Soft Robot

May 2022 – May 2024

- *Department of Mechanical Engineering, Bucknell University, Lewisburg, PA*
- Research Advisor: William Scott, Ph.D.

The research focuses on presenting a new design for a caterpillar-inspired robot with flexible tendon-driven bending segments. The segments are bent in an arc at the neutral configuration, which allows a greater range of motion to lift the feet off the ground than the existing design with all segments flat in the neutral stance. Additionally, a second degree of freedom was introduced to one of the segments to enable steering navigation with controlled out-of-plane motion.

- Developed a new design for a tendon-driven modular 3D printable caterpillar-inspired soft robot with steering capabilities, connecting the segments to the hubs at 45° so that the neutral stance sits at the midpoint of each segment's range
- Designed a steering segment with a fishbone structure thickened so that its bending stiffness matches the standard segment, and showed that turning rate is linear in the steering pulley angle up to a saturation point
- Demonstrated how a computationally simple quasi-static constant-curvature model for planar-legged locomotion can be used in the design of periodic gait control inputs
- Developed a control architecture for the robot using MATLAB and Arduino, and optimized the crawling gait control with the help of computational analysis and simulations
- Demonstrated the crawling locomotion performance of the robot, which involves lifting individual legs off the ground rather than relying completely on sliding motion, in both forward and turning gaits

WORK EXPERIENCE

Engineer Intern

September 2021 – December 2021

- *Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, Guangdong, China*
- Supervisor: Ye Li, Ph.D.
 - Worked at the Major Scientific and Technological Infrastructure for Synthetic Biology as a team member to develop an automatic growth medium coating machine
 - Developed the CAD models of the components, followed up the fabrication process and conducted quality control
 - Assembled the machine, examined its functionality and revised the design to achieve better performance

Programmer Intern

February 2021 – May 2021

- *Xiamen Tumei Software Technology Co., Ltd, Xiamen, Fujian, China*
- Supervisor: Weixing Ji
 - Worked as a team member to develop a smartphone application for an online television charging process
 - Used HTML to design the user interface of the app and collected user feedback to achieve a more user-friendly design

TEACHING AND LEADERSHIP

Course Assistant, 24-351 Dynamics

August 2025 – Present

- *Department of Mechanical Engineering, Carnegie Mellon University, Pittsburgh, PA*
- Supervisor: Victoria Webster-Wood, Ph.D.
 - Led weekly recitation sessions, working through particle and rigid-body dynamics problems with students
 - Held weekly office hours for individual questions on course content and homework
 - Graded homework and exams, and gave written feedback on solution methods
 - Authored and reviewed homework problems and their solutions

Advanced Tutor, Study Group Leader, and New Staff Supervisor

August 2023 – May 2025

- *Teaching and Learning Center, Bucknell University, Lewisburg, PA*
- Supervisor: Katie Daly
 - Facilitated a weekly calculus study group for 5–12 students to reinforce understanding of the class content by summarizing the notes and doing practice problems
 - Held one-on-one calculus tutoring sessions to answer students' questions about class content, homework, and exams

- Supervised new staff at the Teaching and Learning Center, observed their study group or tutor sections and provided feedback

Member, Alumni Relations Committee, Engineering Student Board

August 2023 – May 2025

- *College of Engineering, Bucknell University, Lewisburg, PA*
- Supervisor: Terri Norton, Ph.D.
 - Served as a student voice to the College of Engineering leadership, the faculty, and the Bucknell Engineering Alumni Association
 - Planned and hosted “Happy Hour,” an alumni connection event for students to communicate with alumni and hear about their experiences
 - Operated the Bucknell University College of Engineering LinkedIn Group

Teaching Assistant, ENGR 100: Exploring Engineering

August 2022 – December 2022

- *College of Engineering, Bucknell University, Lewisburg, PA*
- Supervisor: Kelly Salyards, Ph.D.
 - Provided guidance and insight to students during lab sections and held weekly office hours to help students with Arduino and other problems in their project development
 - Guided and helped the students to use needed engineering equipment such as 3D printers, laser cutters, and poster printers

LICENSURE AND CERTIFICATIONS

Fundamentals of Engineering (FE) Exam, Mechanical, NCEES

March 2025

Level 1 Certified Tutor, College Reading and Learning Association (CRLA)

January 2025

Certified SolidWorks Associate (CSWA), Dassault Systèmes

April 2022

TECHNICAL SKILLS

Robotics: ROS | HEBI series-elastic actuator platforms | joint encoder, IMU and torque sensing | Arduino microcontrollers and servo control || hardware experiment design and data collection

Programming: Python | MATLAB/Simulink | Linux | Git

Simulation and Analysis: PyBullet | ANSYS Mechanical | COMSOL | MATLAB symbolic computation

Design and Fabrication: SolidWorks | Fusion 360 | 3D printing | laser cutting | machine shop